

Step-by-Step Guide to Execute EZ-Web / MADSTAR

This guide explains how to use the EZ-Web/MADSTAR stellar evolution interface developed by Richard Townsend. The platform allows users to simulate stellar evolution and generate HR diagrams, luminosity evolution plots, temperature evolution, and stellar structure outputs directly from a web browser.

Official website:

<http://user.astro.wisc.edu/~townsend/static.php?ref=eZ-web>

1. Open the EZ-Web Interface

Open the website in your browser. The interface contains stellar evolution controls and plotting options. No installation is required because the simulations run on the remote server.

2. Select Initial Stellar Parameters

The most important parameter is the stellar mass. Typical values: 0.5 Msun → Low-mass star 1.0 Msun → Sun-like star 5–20 Msun → Massive star You can also modify: Metallicity (Z) Initial helium fraction Mixing parameters Mass-loss options

3. Configure Evolution Controls

Choose the stopping condition for the stellar evolution simulation. Examples: Maximum stellar age Core hydrogen exhaustion Specific evolutionary phase You may also configure: Time resolution Number of models Output frequency

4. Run the Simulation

Press the Run or Execute button. The remote server computes the stellar evolution track. Execution time depends on: stellar mass requested resolution evolutionary duration

5. Analyze the Output

After execution completes, the platform generates plots and downloadable files. Typical outputs include: Hertzsprung–Russell (HR) diagram Luminosity vs time Radius evolution Temperature evolution Internal stellar structure

6. Understanding the HR Diagram

The HR diagram plots luminosity against effective temperature. Important physical interpretation: Main sequence stars evolve from left to right Massive stars evolve faster Red giant phases appear at lower temperatures and

larger luminosities

7. Downloading the Data

Most outputs can be downloaded as text tables or images. Recommended workflow: Download the evolution track
Open in Python, MATLAB, or Mathematica Perform further astrophysical analysis

8. Example Simulation

Example: Mass = 5 Msun Metallicity $Z = 0.02$ Maximum age = $1e8$ years This produces a moderately massive stellar evolution track.

9. Common Problems

Simulation fails:

Reduce resolution or shorten evolutionary duration. **No output generated:**

Check browser compatibility and refresh the page. **Slow execution:**

The remote server may be busy.

10. Scientific Applications

EZ-Web/MADSTAR is useful for: stellar evolution coursework undergraduate astrophysics projects HR diagram analysis mass-luminosity relation studies stellar lifetime estimation

Important Physical Quantities

Quantity	Physical Meaning
Mass	Determines stellar lifetime and evolution speed
Metallicity	Chemical composition of the star
Luminosity	Energy emitted per second
Effective Temperature	Surface temperature
Radius	Physical size of the star

Useful Links: EZ-Web: <http://user.astro.wisc.edu/~townsend/static.php?ref=ez-web> MESA-Web: <http://user.astro.wisc.edu/~townsend/static.php?ref=mesa-web> MESA Documentation: <https://docs.mesastar.org/>